

# UNIX - NETWORK COMMUNICATION UTILITIES

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When you work in a distributed environment then you need to communicate with remote users and you also need to access remote Unix machines.

There are several Unix utilities which are especially useful for users computing in a networked, distributed environment. This tutorial lists few of them –

## The ping Utility

The ping command sends an echo request to a host available on the network. Using this command you can check if your remote host is responding well or not.

The ping command is useful for the following –

- Tracking and isolating hardware and software problems.
- Determining the status of the network and various foreign hosts.
- Testing, measuring, and managing networks.

## Syntax

Following is the simple syntax to use **ping** command –

```
$ping hostname or ip-address
```

Above command would start printing a response after every second. To come out of the command you can terminate it by pressing CNTRL + C keys.

## Example

Following is the example to check the availability of a host available on the network –

```
$ping google.com
PING google.com (74.125.67.100) 56(84) bytes of data.
64 bytes from 74.125.67.100: icmp_seq=1 ttl=54 time=39.4 ms
64 bytes from 74.125.67.100: icmp_seq=2 ttl=54 time=39.9 ms
64 bytes from 74.125.67.100: icmp_seq=3 ttl=54 time=39.3 ms
64 bytes from 74.125.67.100: icmp_seq=4 ttl=54 time=39.1 ms
64 bytes from 74.125.67.100: icmp_seq=5 ttl=54 time=38.8 ms
--- google.com ping statistics ---
22 packets transmitted, 22 received, 0% packet loss, time 21017ms
rtt min/avg/max/mdev = 38.867/39.334/39.900/0.396 ms
$
```

If a host does not exist then it would behave something like this –

```
$ping giiiiiggle.com
ping: unknown host giiiiiggle.com
$
```

## The ftp Utility

Here ftp stands for **F**ile **T**ransfer **P**rotocol. This utility helps you to upload and download your file from one computer to another computer.

The ftp utility has its own set of UNIX like commands which allow you to perform tasks such as –

- Connect and login to a remote host.
- Navigate directories.

- List directory contents
- Put and get files
- Transfer files as ascii, ebcdic or binary

## Syntax

Following is the simple syntax to use **ping** command –

```
$ftp hostname or ip-address
```

Above command would prompt you for login ID and password. Once you are authenticated, you would have access on the home directory of the login account and you would be able to perform various commands.

Few of the useful commands are listed below –

| Command        | Description                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------|
| put filename   | Upload filename from local machine to remote machine.                                                        |
| get filename   | Download filename from remote machine to local machine.                                                      |
| mput file list | Upload more than one files from local machine to remote machine.                                             |
| mget file list | Download more than one files from remote machine to local machine.                                           |
| prompt off     | Turns prompt off, by default you would be prompted to upload or download movies using mput or mget commands. |
| prompt on      | Turns prompt on.                                                                                             |
| dir            | List all the files available in the current directory of remote machine.                                     |
| cd dirname     | Change directory to dirname on remote machine.                                                               |
| lcd dirname    | Change directory to dirname on local machine.                                                                |
| quit           | Logout from the current login.                                                                               |

It should be noted that all the files would be downloaded or uploaded to or from current directories. If you want to upload your files in a particular directory then first you change to that directory and then upload required files.

## Example

Following is the example to show few commands –

```
$ftp amrood.com
Connected to amrood.com.
220 amrood.com FTP server (Ver 4.9 Thu Sep 2 20:35:07 CDT 2009)
Name (amrood.com:amrood): amrood
331 Password required for amrood.
Password:
230 User amrood logged in.
ftp> dir
200 PORT command successful.
150 Opening data connection for /bin/ls.
total 1464
drwxr-sr-x  3 amrood  group    1024 Mar 11 20:04 Mail
drwxr-sr-x  2 amrood  group    1536 Mar  3 18:07 Misc
drwxr-sr-x  5 amrood  group     512 Dec  7 10:59 OldStuff
drwxr-sr-x  2 amrood  group    1024 Mar 11 15:24 bin
```

```

drwxr-sr-x   5 amrood  group      3072 Mar 13 16:10 mpl
-rw-r--r--   1 amrood  group    209671 Mar 15 10:57 myfile.out
drwxr-sr-x   3 amrood  group       512 Jan  5 13:32 public
drwxr-sr-x   3 amrood  group       512 Feb 10 10:17 pvm3
226 Transfer complete.
ftp> cd mpl
250 CWD command successful.
ftp> dir
200 PORT command successful.
150 Opening data connection for /bin/ls.
total 7320
-rw-r--r--   1 amrood  group      1630 Aug  8 1994 dboard.f
-rw-r-----   1 amrood  group      4340 Jul 17 1994 vttest.c
-rwxr-xr-x   1 amrood  group    525574 Feb 15 11:52 wave_shift
-rw-r--r--   1 amrood  group      1648 Aug  5 1994 wide.list
-rwxr-xr-x   1 amrood  group      4019 Feb 14 16:26 fix.c
226 Transfer complete.
ftp> get wave_shift
200 PORT command successful.
150 Opening data connection for wave_shift (525574 bytes).
226 Transfer complete.
528454 bytes received in 1.296 seconds (398.1 Kbytes/s)
ftp> quit
221 Goodbye.
$

```

## The telnet Utility

Many times you would be in need to connect to a remote Unix machine and work on that machine remotely. Telnet is a utility that allows a computer user at one site to make a connection, login and then conduct work on a computer at another site.

Once you are login using telnet, you can perform all the activities on your remotely connect machine. Here is example telnet session –

```

C:>telnet amrood.com
Trying...
Connected to amrood.com.
Escape character is '^]'.

login: amrood
amrood's Password:
*****
*
*
*   WELCOME TO AMROOD.COM
*
*
*****

Last unsuccessful login: Fri Mar  3 12:01:09 IST 2009
Last login: Wed Mar  8 18:33:27 IST 2009 on pts/10

    { do your work }

$ logout
Connection closed.
C:>

```

## The finger Utility

The finger command displays information about users on a given host. The host can be either local or remote.

Finger may be disabled on other systems for security reasons.

Following are the simple syntax to use finger command –

Check all the logged in users on local machine as follows –

```
$ finger
Login      Name      Tty      Idle      Login Time      Office
amrood     pts/0     Jun 25 08:03 (62.61.164.115)
```

Get information about a specific user available on local machine –

```
$ finger amrood
Login: amrood                      Name: (null)
Directory: /home/amrood           Shell: /bin/bash
On since Thu Jun 25 08:03 (MST) on pts/0 from 62.61.164.115
No mail.
No Plan.
```

Check all the logged in users on remote machine as follows –

```
$ finger @avatar.com
Login      Name      Tty      Idle      Login Time      Office
amrood     pts/0     Jun 25 08:03 (62.61.164.115)
```

Get information about a specific user available on remote machine –

```
$ finger amrood@avatar.com
Login: amrood                      Name: (null)
Directory: /home/amrood           Shell: /bin/bash
On since Thu Jun 25 08:03 (MST) on pts/0 from 62.61.164.115
No mail.
No Plan.
```